

GEORGE MARCHMENT

PhD candidate, Paris-Saclay University, LISN Laboratory

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I am a 3rd year PhD candidate studying computer science at Paris-Saclay University. My research interests include data science, graph algorithms, visualisation, AI research and bioinformatics.

EDUCATION

PhD Candidate

2023–present

Paris-Saclay University

Thesis: *A graph-based framework for scientific workflows: modelling, abstraction, and visualisation for reuse*

Supervisors: [Sarah Cohen-Boulakia](#) and [Frédéric Lemoine](#)

Master's Degree in Bioinformatics, Computational Biology Track

2021–2023

[Digicosme scholarship for excellence](#)

Paris-Saclay University

Double Bachelor's Degree in Mathematics and Computer Science

2018–2021

Université Paris-Saclay

PUBLICATIONS

Peer-reviewed international journals

- [J1] **George Marchment**, Sarah Cohen-Boulakia, Alain Denise and Frédéric Lemoine. Workflow abstractions for automatic modularity: improving workflow code reusability. *Future Generation Computer Systems*, 187:108776, February 2027. <https://doi.org/10.1016/j.future.2026.108776> (PDF: <https://hal.science/hal-05737452>)
- [J2] **George Marchment**, Bryan Brancotte, Marie Schmit, Frédéric Lemoine, and Sarah Cohen-Boulakia. BioFlow-Insight: facilitating reuse of Nextflow workflows with structure reconstruction and visualization. *NAR Genomics and Bioinformatics*, 6(3):lqae092, September 2024. <https://doi.org/10.1093/nargab/lqae092>
- [J3] Marine Djaffardjy, **George Marchment**, Clémence Sebe, Raphael Blanchet, Khalid Belhajjame, Alban Gaignard, Frédéric Lemoine, and Sarah Cohen-Boulakia. Developing and reusing bioinformatics data analysis pipelines using scientific workflow systems. *Computational and Structural Biotechnology Journal*, 21:2075–2085, January 2023. <https://doi.org/10.1016/j.csbj.2023.03.003>

Peer-reviewed international conferences

- [C1] Lénora Buggenhoudt, **George Marchment**, Anne-Flore Cabouat, Frédéric Lemoine, Caroline Appert, and Sarah Cohen-Boulakia. MetroFlow: Visualizing Bioinformatics Workflow Source Code as Interactive Metro-Maps. *IEEE Working Conference on Software Visualization (VISSOFT)*, September 2026. <https://hal.science/hal-05737427v1>
- [C2] Marie Schmit, Ulysse Le Clanche, **George Marchment**, Sarah Cohen-Boulakia, Olivier Dameron, Alban Gaignard, Frédéric Lemoine, and Hervé Ménager. A Standards-Based Knowledge Graph that Bridges Scientific Workflows, Run-Time Provenance, and Tool Registries. *Semantic Web Applications and Tools for Health Care and Life Sciences*, March 2026. <https://hal.science/hal-05517640/>

Posters

- [P1] **George Marchment**, Bryan Brancotte, Marie Schmit, Frédéric Lemoine, and Sarah Cohen-Boulakia. (2024) BioFlow-Insight: Leveraging Graph representations to Facilitate the Reuse of Bioinformatics Workflows. Poster presented at the *ACM REP* Conference, Rennes, France. https://gitlab.liris.cnrs.fr/sha-refair/posters/acm-rep-2024/-/blob/main/Poster%20ACM%20REP.pdf?ref_type=heads
- [P2] **George Marchment**, Marie Schmit, Clémence Sebe, Frédéric Lemoine, Hervé Ménager, and Sarah Cohen-Boulakia. (2024) Representing bioinformatics Nextflow workflows in RO-Crate: challenges and opportunities. Poster presented at the *Semantic Web Applications and Tools for Health Care and Life Science* Conference, Leiden, The Netherlands. <https://hal.science/hal-04540040>

Tutorials given

Computational Reproducibility With Scientific Workflows: Analysing viral genomes with Nextflow

July 2025

ACM REP · Tutorial: <https://george-marchment.github.io/acmrep25/>

Book chapters

- [B1] Sarah Cohen-Boulakia, Frédéric Lemoine, **George Marchment**, Marine Djaffardjy, Alban Gaignard, Clémence Sebe, and Khalid Belhajjame. Reuse and Reproduce Bioinformatic Pipelines using Scientific Workflow Systems. In Berlin Universities Publishing, editor, *Workflow systems for large-scale scientific data analysis*. Berlin Universities Publishing, 2026. <https://pasteur.hal.science/pasteur-05640482/>
- [B2] Sarah Cohen-Boulakia, **George Marchment**, Thomas Cokelaer and Frédéric Lemoine. Reprohackathons: Training efforts to increase Bioinformatics Reproducibility using Scientific Workflow systems. In Berlin Universities Publishing, editor, *Workflow systems for large-scale scientific data analysis*. Berlin Universities Publishing, 2026. <https://pasteur.hal.science/pasteur-05640478v1>

TALKS

Peer-reviewed international conferences

BioFlow-Insight: Leveraging Graph representations to Facilitate the Reuse of Bioinformatics Workflows
ACM REP
June 2024

Peer-reviewed national conferences

MetroFlow: automatic, interactive metro-map visualisation for enhancing transparency and comprehensibility of Nextflow workflows
JOBIM
June 2026

Others

Facilitating reuse of Nextflow workflows with structure reconstruction and metro-map visualisation with BioFlow-Insight and MetroFlow
FAIR metadata for bioinformatics software workshop at *ECCB 2026*
September 2026

Workflow abstractions to improve understandability and reusability of workflow code
Visiting the *VIDA Center* (New York University)
April 2026

MetroFlow: Generating Interactive Metro-Maps from Workflow Code
ShareFAIR annual meeting (Pasteur Institute)
Nextflow Summit · Talk: <https://summit.nextflow.io/2025/virtual/agenda/metroflow-generating-interactive-metro-maps-from-workflow-code/>
December 2025
October 2025

Facilitating the reuse of scientific workflows with structure abstraction
Visiting the *Software Sustainability Institute* (University of Manchester)
June 2025

BioFlow-Insight: facilitating reuse of Nextflow workflows with structure reconstruction and visualisation
ShareFAIR annual meeting (Pasteur Institute)
Team seminar (Paris-Saclay University, LISN)
November 2024
January 2025

Initial findings on workflow process similarity
ShareFAIR kickoff (Paris-Saclay University)
November 2023

SOFTWARE

MetroFlow

Website: <https://metroflow.pasteur.cloud/> · Source code: <https://gitlab.pasteur.fr/sharefair/metroflow>

BioFlow-Insight

Website: <https://bioflow-insight.pasteur.cloud/> · Source code: <https://gitlab.liris.cnrs.fr/sharefair/bioflow-insight>

Github-Crawler

Source code: <https://gitlab.liris.cnrs.fr/sharefair/github-crawler>